Provenance of the atomic correlation constant $a_{\rm corr}$ in the substrate framework: what is derived, what is conditional, and what remains open

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This is a provenance record for the correlation constant $a_{\rm corr}$ as the substrate corpus treats it, so that every statement can be checked against a file, a function and a commit. In the corpus $a_{\rm corr} = \epsilon_2 - \eta_2$ is the high-nuclear-charge, same-centre, two-electron second-order correlation coefficient of the Coulomb comparison model. $\eta_2 = \frac{9}{32}\ln\frac{3}{4} - \frac{13}{432}$ is derived in closed form; ϵ_2 is a published value, not a substrate result, so the number $a_{\rm corr} = -0.046663253999494962615$ is a diagnostic. The corpus writes $a_{\rm corr}$ in two parts in several ways: a contact block fixed by the coincidence limit plus a non-contact remainder $\mathcal{R}_{\rm tail}$; an s-wave part (30.7 percent) plus a higher-partial-wave part (69.3 percent), of which the coincidence limit fixes only the large-l tail; and a charge-tail (single-source response) part plus a two-centre correlation part. The corpus builds $a_{\rm corr}$ from both of the framework's fixed inputs: the Born-Infeld Lagrangian and the closed-form particle expressions. The particle expressions enter through the electron mass that sets the hartree and the Bohr radius, through the coupling that converts to electronvolts (conditional: the particle module marks its a_0 as not forward-closed), and through the Born-Infeld core scale relative to the atom; they do not, on the present record, supply the dimensionless value. The August element-generator route that called $a_{\rm corr}$ forward used a Coulomb-model Jastrow estimate of a differently defined quantity, installed the literature value 0.04667, and fitted its occupation shape; its under-one-percent period-2 closure was superseded in September. None of the splits determines $a_{\rm corr}$'s value from these premises. The test-charge kernel is not the equal-source interaction (exact contact ratio $4(\sqrt{2}-1)/3$). The two-centre part rests on the static two-source Born-Infeld energy. Its frozen superposition trial is closed in the sense of an exact linear differential equation with an exact algebraic-period germ, so it is computable by series continuation; the true static minimum is fenced but has no series; and the identification of either with the physical correlation of moving modes is a registered commitment that is not discharged. The physical $a_{\rm corr}$ remains open. We list the routes tried and the reasons each does not close, with their sources.

I. PURPOSE AND SCOPE

The owner's summary of this line of work is that $a_{\rm corr}$ "was derived into two parts with $R_{\rm tail}$ closed as a series". This paper sets out what the corpus actually establishes about that statement, with the evidence for each step, and reports where the statement aligns, partly aligns, or conflicts with the record. The owner's correction adds that $a_{\rm corr}$ "is not fixed merely from the lagrangian. It is also built based on the fixed closed form expressions for the particles"; Sec. IV traces how. It is a provenance document: it introduces no new derivation, and every number is a macro generated by `papers/two_center_born_infeld/generate_numbers.py` from a corpus receipt at a recorded commit, or recomputed from a closed form. That generator also verifies that each of the 60 formula strings transcribed into the two companion papers occurs verbatim in its cited source, and that each cited function exists. The references are origin/dev at `830cc5170df5`, the frozen-ODE branch at `3b8c2653996d`, the independent order-17 check at `cdd0f70ffa18`, the bond-research branch at `367287a67291` and the a_corr lane branch at `20cb3828b355` (full hashes in `papers/two_center_born_infeld/provenance_refs.json`). The static two-centre problem is treated in full in the companion paper [1]; here it appears only as one part of $a_{\rm corr}$.

A. Standing context

The substrate framework is a Born-Infeld Lagrangian [2] describing an elastic solid with an excited $E_8 \times E_8$ dyad. $E_8 \times E_8$ is observationally overdetermined, D_{16} is excluded (`book/gr_qm_unification/sympy/substrate_theorems/rank16_observational_overdetermination.py`), and that includes its twist and spinor sector, its rank eight and its D_8 map. The framework has forward-derived quantization, entanglement and relativistic time dilation and is

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not classical. The particle mass equations are fixed closed forms in `book/scripts/substrate_mass_spectrum.py`. Particles are localized rotating wave modes of finite radius: two electrons in an atom are two lasting ripples of one medium, and their correlation is the discount in the energy of their joint, interfering pattern relative to the sum of the separate patterns. The Coulomb comparison model used below to define $a_{\rm corr}$ is an imported reference problem, not the substrate's account of that discount.

B. Status vocabulary

DEFINITION; DERIVED (forward from stated premises; the premises are named); CONDITIONAL (rests on a registered commitment); CERTIFIED-EXACT; DIAGNOSTIC (a computation inside an imported comparison model or with literature inputs); LITERATURE (an external value or theorem); OPEN; WITHDRAWN (asserted earlier, later retracted in the corpus). Table II gives each claim's source file, function where one exists, and the last commit touching it; a dagger marks a commit not contained in origin/dev.

II. WHAT $a_{\rm corr}$ IS IN THE CORPUS

A. Definition

For two electrons bound to one nucleus of charge Z in the nonrelativistic Coulomb problem, the ground energy has the large-Z expansion $E(Z) = -Z^2 + \frac{5}{8}Z + \epsilon_2 + O(1/Z)$ in hartree, and the Hartree-Fock energy has the same first two terms and second-order coefficient η_2. The corpus defines

$$a_{\rm corr} = \epsilon_2 - \eta_2 \tag{1}$$

(`atomic_correlation_constant_not_forced.py::verdict`; DEFINITION). The $1/Z$ expansion and the Hartree-Fock reference are perturbative constructs of the imported Coulomb model; in the substrate framework they serve only as a comparison problem, under the quarantine of perturbation theory as a method, and a substrate-native correlation must come from the moving-mode energy functional of Sec. V and the open items. It is a one-centre constant: it carries no dependence on a separation between centres, so it is not the two-centre correlation energy that a bond depth needs. The $O(Z)$ terms are equal because the exchange-lowered spatially antisymmetric partner of $1s^2$ is excluded in both curves (`pauli_shared_oz_term`), and the static monopole content $J = \frac{5}{8}Z$ cancels in the difference.

B. The closed part: η_2

The first-order orbital equation $(-\frac{1}{2}\nabla^2 - 1/\rho + \frac{1}{2})\phi_1 = -(V_C - \frac{5}{8})\phi_0$ with $V_C(r) = (1 - (1+r)e^{-2r})/r$ is solved by $\phi_1 = f\phi_0$ in the Dalgarno-Lewis reduction [3], which turns it into a first-order equation for $w = f'$ whose solution regular at the origin is elementary. The energy integral then gives

$$\eta_2 = \frac{9}{32}\ln\frac{3}{4} - \frac{13}{432} = -0.111003175469655978435, \tag{2}$$

the logarithm arising by Frullani integration of the $1/r$ part of f' (`eta2_symbolic_derivation`; establish-or-bail checks of the defining equation, regularity, decay and the self-adjoint identity). It agrees with the published closed form of Loos and Gill [4], which is an oracle here. Status: DERIVED, inside the imported Coulomb operator, whose kinetic and potential terms are not derived from the Born-Infeld medium in this calculation.

C. The open part: ϵ_2

The corpus value $\epsilon_2 = -0.15766642946915094105$ is read from the publication of Wang [5] (`second_order_partial_waves.py::published_eps2`; LITERATURE). Hence the corpus number

$$\begin{aligned} a_{\rm corr} &= -0.15766642946915094105 - (-0.111003175469655978435) \\ &= -0.046663253999494962615, \qquad |a_{\rm corr}| = 0.0466632539995, \end{aligned} \tag{3}$$

is a DIAGNOSTIC that inherits the provenance of the published ϵ_2; it is not a substrate prediction. The corpus reduces ϵ_2 to a sum over total-$L = 0$ partial-wave channels. The s-wave particular solution $\frac{5}{8}r_> - \frac{3}{8}\ln r_> + \frac{3}{16r_>}$ is exact; the large-l tail follows the Schwartz law $-\frac{45}{256}(l+\frac{1}{2})^{-4} + \frac{225}{1024}(l+\frac{1}{2})^{-6} + \dots$ [6], whose sum is elementary through half-integer Hurwitz values; the matching data at the crossover shell $r_1 = r_2$ are fixed in closed form by a field junction. The crossover correction, however, is not in the five-dimensional elementary-plus-exponential-integral homogeneous space of the channel operator, so the s-wave channel function is not of that elementary form and $\epsilon_2(0)$ does not close by this assembly; numerically $\epsilon_2(0) = -0.125334(1)$ (`eps2_swave_crossover_is_non_elementary`). This is a non-membership result for one assembly, not an impossibility proof for a closed form. Status of a closed form for ϵ_2: OPEN.

III. THE TWO-PART DECOMPOSITIONS

The corpus writes $a_{\rm corr}$ as two parts in three different senses. They must not be conflated, and the symbol $R_{\rm tail}$ carries different meanings across them (written $\mathcal{R}_{\rm tail}$ here for the remainder of the first split).

A. Contact block plus non-contact remainder

Write the target as a global integral $a_{\rm corr} = \int_0^\infty G(r_{12})\, w(r_{12})\, dr_{12}$ of a two-centre field-energy density against a pair measure, choose an overlap radius r_*, and split

$$a_{\rm corr} = \sum_k \mu_k G_k + \mathcal{R}_{\rm tail}, \qquad \mu_k = \int_0^{r_*} r^{p_k} w\, dr, \qquad \mathcal{R}_{\rm tail} = \int_{r_*}^\infty G\, w\, dr, \tag{4}$$

where $G(r) = \sum_k G_k r^{p_k}$ near coalescence (`acorr_proximity_cap_rank_analysis.py`). Matching the global density to the local coincidence hole, $G_k = L_k$, is a square system: one equation and one open contact coefficient per order, with no surplus equation, so the match transfers local moments into contact data and places no constraint on the global integral (`substrate_correlation_coincidence_match_underdetermines`; DERIVED). A Born-Infeld proximity cap at a minimum separation is a contact-local boundary condition: symbolically and over 200 random instances its row lies in the span of the match rows, its projection on the $(a_{\rm corr}, \mathcal{R}_{\rm tail})$ columns vanishes, and the $(a_{\rm corr}, \mathcal{R}_{\rm tail})$ subsystem stays rank one in two unknowns; the cap's consistency fixes the minimum separation, not $a_{\rm corr}$ (`substrate_correlation_proximity_cap_is_contact_local_no_rank`; DERIVED). This split is a bookkeeping identity; it shows that no local condition reaches $a_{\rm corr}$, and it names $\mathcal{R}_{\rm tail}$ as the residual carried by the global two-centre field.

B. s-wave plus higher partial waves

With literature per-l increments of the Coulomb problem (Carroll, Silverstone and Metzger [7]) the corpus finds the s-wave correlation $\epsilon_2(0) - \eta_2 = -0.014326825$, 30.7 percent of $a_{\rm corr}$, and 69.3 percent in $l \geq 1$ (`acorr_partialwave_decomposition.py`; DIAGNOSTIC). The coincidence limit $r_{12} \to 0$ is the large-l limit: it reproduces the shared cusp law, whose sum from $l = 0$ is $+11.23$, nowhere near ϵ_2, while most of $a_{\rm corr}$ sits in the low-l head that the coincidence data never reach. The resisting content is the short-range Kato coalescence cusp [8], whose branch point gives the partial-wave expansion its power-law tail.

C. Charge-tail part plus two-centre correlation

a. Charge tail. The single-source Born-Infeld potential and the linear response of an infinitesimal test charge are evaluated in `acorr_charge_tail_dependency_FINDINGS.md`. For $s \geq 0$ and core scale $r_0 > 0$, the substitutions $r = r_0(1-t)/t$, $t = au$, $a = r_0/(s+r_0)$ give

$$U(s) = \frac{1}{s+r_0}\int_0^1 \frac{du}{\sqrt{(1-au)^4 + (au)^4}}, \qquad \frac{1}{s} - \frac{r_0^4}{10 s^5} \leq U(s) \leq \frac{1}{s}, \tag{5}$$

with the quartic in $[1/8, 1]$ on the fixed interval, which removes a scale-dependent quadrature failure (DERIVED, scalar route only; landed numerical-domain component). The same record states the scope plainly: the mechanical

displacement, strain, scalar potential and conjugate flux are distinct types, a scalar $1/r$ potential does not identify a compatible mechanical strain, and no physical $a_{\rm corr}$ follows. The decisive check is at contact. By displacement-preserving dilation a source of charge q has energy $Kq^{3/2}$ with $K = \Gamma(1/4)^2/(6\sqrt{\pi}) = 1.236049784867581$, so the interaction of two finite sources at contact is $V(q_1, q_2; 0) = K[(q_1+q_2)^{3/2} - q_1^{3/2} - q_2^{3/2}]$, while the test-charge coefficient is its one-sided derivative. For two unit sources

$$\frac{V(1,1;0)}{U(0)} = \frac{4(\sqrt{2}-1)}{3} = 0.5522847498307934 \neq 1, \qquad U(0) - V(1,1;0) = \Big(\frac{7}{2} - 2\sqrt{2}\Big)K = 0.8300975079803747, \quad (6)$$

(`bi_test_charge_pair_energy.py::contact_interaction_certificate`; CERTIFIED-EXACT). The isolated-source test-charge potential is therefore not the equal-source interaction already in the static point-source diagnostic, and the partial-wave calculations that used it as a pair kernel are comparison calculations whose transfer to two finite sources is not established. The static linear far-field strain of a spherical source is pure longitudinal and pure monopole; inside the Coulomb comparison operator its content is the $l = 0$ direct term that cancels in $a_{\rm corr}$, and the retarded magnetic wake enters at $O(\alpha^2) \sim 5 \times 10^{-5}$ (`substrate_correlation_strain_tail_and_wake_not_the_resisting_piece`; DIAGNOSTIC). This does not exclude wake, twist or retardation contributions to the physical energy of finite moving modes; the framework holds that those effects are always present, and their size must be computed, not assumed.

b. Two-centre correlation. The second part is the energy of the joint two-source field, which is where Born-Infeld nonsuperposition acts. The corpus registers its identification with $a_{\rm corr}$ as the commitment `rtail_two_center_m11_physical_identification`: mapping the static Euclidean two-centre energy to the correlation of moving atomic modes requires the finite moving-mode reduction (observer clock and measure, retarded wake, angular momentum and twist, interference), which is not built (CONDITIONAL). The M11 route validation, which tried to compute $a_{\rm corr}$ from the true two-electron field minimizer, returns open with that blocking antecedent named, `full_elastic_observer_energy is None`, together with the missing physical two-electron bound mode; it also records that softening the interaction alone, with the nonrelativistic framework fixed, shifts $a_{\rm corr}$ by $O(r_0^2) \sim 10^{-9}$ at atomic scale, so the leading part reduces to the Coulomb $\epsilon_2 - \eta_2$ unless the moving-mode terms change it (`acorr_m11_true_minimizer_route_validation.py`; OPEN).

IV. HOW THE CLOSED-FORM PARTICLE EXPRESSIONS ENTER, AND THE ELEMENT-GENERATOR ROUTE

The owner's correction is that $a_{\rm corr}$ "is also built based on the fixed closed form expressions for the particles". The corpus confirms that the particle equations of `book/scripts/substrate_mass_spectrum.py` enter the $a_{\rm corr}$ calculations, and the element-generator scripts of August 2026 present a route that they call forward. This section traces each step to its source, with its status. All steps below sit inside the same nonrelativistic Coulomb comparison problem as Sec. II unless stated otherwise.

a. Step P1: the electron mass. Under the module's default anchor (`default='structural'`), the electron rest energy is the fixed closed form on the ϕ/E_8 ladder times the structural scale $h_{\rm eff}$ (`substrate_mass_spectrum.py::derive_constants`). The electron mass sets the hartree and the Bohr radius, the units in which $a_{\rm corr}$ is a pure number. Status: FIXED CLOSED FORM, a framework premise; it fixes the unit, not the dimensionless value $a_{\rm corr}$.

b. Step P2: the coupling and the energy unit. The element generator converts hartree to electronvolts with $E_h = \alpha_{\rm sub}^2 m_e c^2$, $\alpha_{\rm sub} = \lambda_{C,e}/(2\pi a_0)$, taken from the same module (`forward_p_row_generator.py::hartree_eV`); if that import fails it silently falls back to the measured hartree. The module's own status block marks a_0 and the Rydberg energy as not forward-closed: they inherit an α that consumes imported QFT corrections, with the Born-Infeld $\alpha(0)$ closure pending. Status: CONDITIONAL. This step sets the eV value of $a_{\rm corr}$, not its value in hartree.

c. Step P3: the Born-Infeld core scale relative to the atom. The size of the Born-Infeld softening of the pair interaction relative to the Coulomb value is set by the core radius r_0 in Bohr units, which uses the identification of the single-source energy with the electron rest energy. The M11 validation records that softening alone shifts $a_{\rm corr}$ by $O(r_0^2) \sim 10^{-9}$ at atomic scale (Sec. III C). Status: CONDITIONAL on that identification and on the fixed nonrelativistic framework.

d. Step P4: the r_{12} Jastrow route. `derive_pair_correlation_constant_forward.py` (with `derive_helium_correlation_forward.py`) evaluates the point-Coulomb Hamiltonian, `pot = -Z / r1 - Z / r2 + 1.0 / r12` with the Schrödinger kinetic operator, for the trial $e^{-\zeta(r_1+r_2)}(1 + cr_{12})$ by fixed-seed variational Monte Carlo, defines $E_{\rm corr}(Z) = -(Z - 5/16)^2 - \min E(\zeta, c; Z)$, and extrapolates in Z. Its recorded values are $Z = 2$: 0.0435, $Z = 4$: 0.0467, $Z = 8$: 0.0472, $Z = 16$: 0.0497 hartree; the script's own least-squares extrapolation $a + b/Z$ over $Z = 4, 8, 16$ gives 0.04995, consistent with its commit message ("extrapolating to ∼0.05 Ha"), while `FORWARD_ELEMENT_GENERATOR.md` reports $a \approx 0.047$. No particle expression and no Born-Infeld term enters this calculation; its "tension cusp" is the

Kato cusp of the Coulomb operator. Its reference is the screened single-exponent energy, whose Z^0 term is $-25/256 = -0.09765625$, not the Hartree-Fock η_2; relative to that reference the exact Z^0 correlation is $|\epsilon_2 + 25/256| = 0.0600102$, so the one-parameter Jastrow estimate is a variational partial recovery of a differently defined quantity, and its closeness to $|a_{\rm corr}| = 0.0466632539995$ does not identify the two. The check passes when the result lies within 0.008 of the fitted generator value 0.04738; Monte Carlo uncertainty is not quantified. Status: DIAGNOSTIC; nothing in it is fitted, but it is neither substrate-native nor the defined quantity.

e. Step P5: the value actually installed. The element generators install the literal 0.04667 (`forward_missing_terms_generator.py`; `LEGACY_DIAGNOSTIC_A_CORR=sp.Rational(4667,100000)` in `forward_generator_complete.py`). That number is the He-like high-density limit, the corpus diagnostic $|\epsilon_2 - \eta_2|$ of Sec. II, whose ϵ_2 is a published value. Status: LITERATURE VALUE INSTALLED; the current files label it a legacy diagnostic.

f. Step P6: the occupation shape and the 3:4 ratio. The missing-terms polynomial $a\,k(3k-11)/8$ with $b/a = -3/4$ was obtained by a least-squares fit of the six period-2 residuals (`forward_missing_terms_generator.py`: "This script DISCOVERS the RATIO pattern (fits to expose the 3:4 lock and the single constant a)"). The identification of 3/4 with the spin-$\frac{1}{2}$ Casimir used in the particle sector is recorded as a lead "TO CONFIRM", and the particle-generator origin of the value 0.0467 as open. The September record rejects 1/21 and 3/64 as exact values of $|a_{\rm corr}|$: they differ from it by +2.05 and +0.45 percent, outside its error bar. Status: shape and ratio FITTED; particle-sector origin OPEN.

g. Step P7: the September corrections. The claim that this generator plus one constant closes period 2 to under one percent with zero fits was superseded in September 2026. The legacy optimizer minimized a removed-orbital expression, not the atom's common occupied energy; the exact derivative defects are $[0, -21/64, -63/64, -585/256, -921/256, -1341/256]$ for $k = 1, \ldots, 6$ (`atomic_generator_energy_consistency_FINDINGS.md`), and selecting the common energy fixes that objective (`atomic_p_relaxed_energy_FINDINGS.md`); the two correction commits are in Table III. The candidate shape is also not what a single-amplitude pair correction produces (`atomic_pshell_pair_structure_FINDINGS.md`). A probe by the ionization-generator lane (`40cd2c13f73b`, unreviewed) reproduces the September table and gives B to Ne at −16.1 to −53.6 percent with the corrected objective, and finds that the fitted shape does not transfer to the corrected generator. Status: the under-one-percent closure is WITHDRAWN.

h. Relation to the decomposition of Sec. II and to M11. The installed number and the corpus definition are the same quantity, $a_{\rm corr} = \lim_{Z\to\infty}(E_{\rm exact} - E_{\rm HF}) = \epsilon_2 - \eta_2$, whose value is known numerically from published work. The Jastrow route is a different route to a related but differently defined quantity of the same Coulomb model, consistent in size and not in conflict with the September identification, which kept the stored value and found no closed form. Both routes sit inside the Coulomb comparison problem, so both lack the M11 step: the moving-mode two-electron energy functional of the Born-Infeld medium. The particle equations fix the units and the atomic scale through which that functional would be expressed; on the present record they do not supply the dimensionless value of $a_{\rm corr}$.

V. $R_{\rm tail}$ AND ITS SERIES

In the files named `rtail_*` the symbol $R_{\rm tail}(d)$ denotes the static two-source Born-Infeld energy at separation d, either of the true minimizer $E_{\rm min}(d)$ or, in older files, of the frozen superposition trial $W_{\rm sup}(d)$ (the reduction once labelled the true $R_{\rm tail}$ was relabelled as the frozen trial; see the timeline). The companion paper [1] gives the full account; the results that bear on the owner's statement are these.

1. The frozen trial $W_{\rm sup}$ satisfies an exact linear differential equation of order 18 with integer polynomial coefficients, certified by exact divergence identities (replayed for this record), and a smaller order-17 annihilator has independent source clearance on its branch. The physical branch has order at least 4 and is not algebraic. An exact germ of eighteen algebraic periods at $d = 1$ selects the solution, so $W_{\rm sup}$ is computable to arbitrary precision by series continuation once the germ is enclosed to that precision; a recorded continuation step agrees with independent quadrature to 1.07×10^{-81}, and an unreviewed multi-step continuation to 2.1×10^{-53} or better on $1.25 \le d \le 4$ (observed agreements, not error bars). In this precise sense the frozen trial is "closed as a series".

2. The frozen trial is a strict upper bound on the static energy, never the energy itself. For the true minimum the corpus has certified fences (the contact force 1/2 from a contact enclosure, the interval $0.63726946 \le V(1) \le 0.64626656$, a dual lower bound) but no differential equation and no series; its large-separation law is conditional on a registered commitment.

3. Neither static object has been identified with the physical two-mode correlation (the commitment above), and neither is the remainder $\mathcal{R}_{\rm tail}$ of the first split, which is an integral of a density that the corpus has not constructed.

The former claim that the physical two-centre energy is a proved non-elementary order-four Picard-Fuchs period is WITHDRAWN: `rtail_known_problems_context.md` states that the former claim is withdrawn and that the checked algebra concerns the frozen trial. The derivations README at the recorded origin/dev commit still carried the older "now PROVEN NON-ELEMENTARY" paragraph; the corpus commit that accompanies this paper replaces it with the current status, and corrects the two-centre synthesis module, which had called the true large-separation law exact.

VI. ROUTES TRIED AND THEIR OUTCOMES

TABLE I: Routes to a_{corr} recorded in the corpus.

route	outcome	source
Constant installed in the forward element generator (August)	the original 2p diagnostic installs 0.04667 hartree as an input; its forward provenance was not established, and a later commit separated it from the cross-shell correlation E_c	a_corr lane journal; timeline
ϵ_2 by Green's-function assembly	crossover correction outside the elementary plus Ei span; $\epsilon_2(0)$ does not close by this assembly (non-membership)	`eps2_swave_crossover_is_non_elementary`
Rigid limit versus deformed metric	the lapse rationalizes the source but the acoustic-measure root cancels it; the universal cancellation was later withdrawn and the observer measure must be derived	`substrate_correlation_hole_is_a_deformed_metric_problem`
Global field-energy route	the saturation root resums globally and the single-centre building blocks are $\Gamma(1/4)$-clean; the correlation functional and its field-energy map remain open	`substrate_correlation_global_field_energy_route`
Coincidence match	square system; under-determines a_{corr}	`substrate_correlation_coincidence_match_underdetermines`
Static strain tail and wake	inside the comparison operator: the static tail is $l = 0$ and cancels, the wake is $O(\alpha^2)$; not a statement about finite moving modes	`substrate_correlation_strain_tail_and_wake_not_the_resisting_piece`
Cuspless test-charge kernel	comparison calculation; the kernel is not the equal-source interaction	`substrate_native_cuspless_correlation_constant`
Proximity cap	contact-local, adds no rank	`substrate_correlation_proximity_cap_is_contact_local_no_rank`
Order-four Picard-Fuchs closure of R_{tail}	withdrawn; downgraded to a rank-six upper bound with open obligations, then superseded by the exact order-18 equation for the frozen trial	`rtail_known_problems_context.md`
M11 true-minimizer route	open; the moving-mode two-electron functional and the physical bound mode are missing	`acorr_m11_true_minimizer_route_validation.py`
Element-generator r_{12} Jastrow route (August)	Coulomb-model VMC of a screened-reference quantity, extrapolated near 0.05; generator installs the literature 0.04667 and a fitted shape; under-one-percent closure withdrawn in September	Sec. IV
Harmonium pair law	not a forced universal constant	`harmonium_correlation_not_forced.py`

The module title `atomic_correlation_constant_not_forced.py` records the cumulative result: its verdict exports `"a_corr": None` and `"physical_generator_complete": False`. "Not forced" there means that no route recorded so far forces a value; it is not a proof that no route can.

VII. WHAT IS CERTIFIED, AND TO WHAT PRECISION

- Exact: η_2 in closed form; the contact ratio $4(\sqrt{2}-1)/3$; the endpoint energies $2K$ and $2^{3/2}K$ of the static two-source problem; the order-18 annihilator of the frozen trial and its indicial data; the frozen large-separation law $W_{\mathrm{sup}} = 2K + 1/d - (3K/4)/d^4 + o(d^{-4})$; the lower bound 4 on the frozen trial's order and its nonalgebraicity.

- Interval-certified: the true static minimum's contact enclosure on $0 \le d \le 1/2$ and its interval at $d = 1$.
- Observed (agreement of routes, not error bars): the frozen trial's values and continuation agreements; the numerical $\epsilon_2(0)$.
- Literature inputs: ϵ_2, the per-l increments used in the partial-wave split, and the generator literal 0.04667.
- Fitted: the period-2 occupation shape and its 3:4 ratio. Withdrawn: the period-2 under-one-percent closure.
- Fixed closed forms entering as units and scale: the electron mass (premise); the coupling for the eV conversion (conditional).

VIII. ASSESSMENT OF THE OWNER'S STATEMENT

The statement partly aligns with the record. It aligns in that the corpus does split a_{corr} into two parts, in each of the three senses above, and in that the frozen two-centre trial energy is closed in the sense of an exact differential equation with an exact germ, computable to arbitrary precision by series continuation. It conflicts in three places. First, a_{corr} is built from both the Lagrangian and the closed-form particle expressions, as the owner states, but neither input, nor any split, determines its dimensionless value on the present record (Sec. IV): the contact-plus-remainder split is a bookkeeping identity whose remainder is unconstructed, the partial-wave split uses published Coulomb values, and the charge-tail-plus-two-centre split rests on an identification that is a registered, undischarged commitment. Second, what is closed as a series is the frozen superposition trial, an upper bound, not the true static minimum and not the correlation of moving modes. Third, the corpus value of a_{corr} inherits the published ϵ_2; it is a diagnostic, not a substrate result. The physical a_{corr} is open.

IX. WHAT REMAINS OPEN

1. The physical moving-mode two-electron correlation functional, with its observer clock and measure (by the owner's ruling, one global tick and the actual local voxel volume), retarded strain field, motion wake, twist and interference, and its identification with a_{corr}; and the physical bound mode it requires.
2. The charge sector of the master Lagrangian's cross term (phase-carrying mode class, source coupling, $1/r$ tail and Maxwell regime), which the a_corr lane orders before the bound mode.
3. Whether the particle sector fixes the dimensionless value of a_{corr} (the 3/4 Casimir lead and the origin of the number), and a forward-closed coupling for the eV conversion.
4. A closed form, or a proof of impossibility, for ϵ_2 and for $\epsilon_2(0)$.
5. The remainder $\mathcal{R}_{\mathrm{tail}}$ as an actual integral of a constructed density.
6. For the static two-centre energy: the items listed in [1], including the minimal order of the frozen trial and any series or equation for the true minimum.
7. Verification of corpus citation fields not checked for this record, for example the forcing node's statement of what Carley and Kiessling establish about the linear Bopp-Podolsky potential.

Appendix A: Claims, status and sources

Paths are relative to `book/gr_qm_unification/sympy/`.

TABLE II: Claims of this paper with status and source.

id	claim	status	source	commit
A1	a_corr is defined as eps_2 - eta_2 of the Coulomb comparison model (high-Z, same centre, two electrons)	DEFINITION	`substrate_theorems/atomic_correlation_constant_not_forced.py::verdict`	`56d7c16cfe`

id	claim	status	source	commit
A2	eta_2 = (9/32)ln(3/4) - 13/432 by the Dalgarno-Lewis reduction	DERIVED (imported Coulomb operator)	substrate_theorems/atomic_correlation_constant_not_forced.py::eta2_symbolic_derivation	56d7c16cfe
A3	eps_2 = published value (Wang 2013), not a substrate result	LITERATURE	imported_theorems/second_order_partial_waves.py::published_eps2	56d7c16cfe
A4	s-wave and l>=1 split of a_corr from literature per-l increments	DIAGNOSTIC (literature inputs)	derivations/acorr_partialwave_decomposition.py	674a28b5fd
A5	the s-wave crossover correction lies outside the elementary plus Ei span (non-membership only)	DERIVED (conditional)	substrate_theorems/atomic_correlation_constant_not_forced.py::eps2_swave_crossover_is_non_elementary	56d7c16cfe
A6	coincidence match is a square system; it fixes contact data, not a_corr	DERIVED	substrate_theorems/atomic_correlation_constant_not_forced.py::substrate_correlation_coincidence_match_underdetermines	56d7c16cfe
A7	proximity cap adds zero rank; (a_corr, R_tail) block stays rank 1 in 2 unknowns	DERIVED	derivations/acorr_proximity_cap_rank_analysis.py	56d7c16cfe
A8	test-charge contact ratio V(1,1;0)/U(0) = 4(sqrt2-1)/3, not 1	CERTIFIED-EXACT	substrate_theorems/bi_test_charge_pair_energy.py::contact_interaction_certificate	b436d403d6
A9	scalar point-source potential: exact substitution, quartic range [1/8,1], Coulomb bracket	DERIVED (scalar route only)	derivations/acorr_charge_tail_dependency_FINDINGS.md	cfa43b6430
A10	static linear strain tail is l=0 and the wake is O(alphaˆ2) inside the Coulomb comparison only	CONDITIONAL DIAGNOSTIC	substrate_theorems/atomic_correlation_constant_not_forced.py::substrate_correlation_strain_tail_and_wake_not_the_resisting_piece	56d7c16cfe
A11	static two-centre energy to a_corr identification	COMMITMENT (M11-gated)	substrate_theorems/rtail_two_center_synthesis.py::two_center_synthesis	56d7c16cfe
A12	M11 true-minimizer route: physical moving-mode functional and bound mode missing	OPEN	derivations/acorr_m11_true_minimizer_route_validation.py	56d7c16cfe
A13	former order-4 Picard-Fuchs non-elementarity claim	WITHDRAWN	derivations/rtail_known_problems_context.md	745293d9c5
A14	physical a_corr value from the framework's premises (Lagrangian and particle equations)	OPEN	substrate_theorems/atomic_correlation_constant_not_forced.py::verdict	56d7c16cfe
A15	electron mass from the fixed closed-form particle equations (default structural anchor)	FIXED CLOSED FORM (framework premise)	book/scripts/substrate_mass_spectrum.py::derive_constants	d90bb8280a
A16	energy unit alphaˆ2 m_e cˆ2 that expresses a_corr in eV, from the particle module	CONDITIONAL (module marks a_0 and Ry not forward-closed)	book/scripts/forward_p_row_generator.py::hartree_eV	de463350ee
A17	r12 Jastrow VMC 'forward a': Coulomb Hamiltonian, screened reference, extrapolated	DIAGNOSTIC (Coulomb model; not the defined quantity)	book/scripts/derive_pair_correlation_constant_forward.py::corr_at_Z	79ab032f5a
A18	b/a = -3/4 identified with the spin-1/2 Casimir of the particle sector	OPEN (lead, to confirm)	book/scripts/FORWARD_ELEMENT_GENERATOR.md	7fd6e5c283
A19	element generator installs a_corr = 4667/100000 as a legacy diagnostic literal	LITERATURE VALUE INSTALLED	book/scripts/forward_generator_complete.py::legacy_orbital_ie_diagnostic	82ead9b8d9
A20	occupation shape k(3k-11)/8 and b/a = -3/4 found by least-squares fit of the period-2 residual	FITTED	book/scripts/forward_missing_terms_generator.py	c0610401eb

id	claim	status	source	commit
A21	legacy optimizer minimized a removed-orbital expression, not the atom's energy (September correction)	CERTIFIED-EXACT (defect)	derivations/atomic_generator_energy_consistency_FINDINGS.md	de463350ee
A22	period 2 under 1% with zero fits superseded; corrected objective gives B to Ne -16.1 to -53.6 percent	UNREVIEWED (other lane probe)	work_journal/ionization-generator-period3-open.md	40cd2c13f7†

Appendix B: Dated history

Commits are found by exact subject in the remote history; times are UTC; a dagger marks a commit not contained in origin/dev.

TABLE III: The history of a_{corr} and R_{tail} in the corpus.

UTC	commit	subject
2026-08-11 13:52Z	7a4f007840	feat(abi): forward helium correlation -- the r12 term closes the mean-field residual to +0.42%
2026-08-11 18:18Z	79ab032f5a	feat(book): forward-derive the missing-terms constant a = pair-correlation energy
2026-08-11 18:23Z	74bebdae80	feat(book): the COMPLETE forward generator ; zero fitted constants, 2p row <1%
2026-08-11 21:27Z	a4dc3e19eb	fix: correct the E_c=a_corr over-claim ; verified they are different quantities
2026-09-17 02:30Z	b2a7b01ba6	feat(gr_qm): a_corr is measured, not derivable -- and the corpus disagrees with itself by 2
2026-09-18 01:32Z	1ae99b7475	feat(gr-qm): eps_2(0) does not close by the Green's-function assembly -- the crossover correction is non-elementary
2026-09-18 05:31Z	5fbbc005ff	research(gr-qm): prove the rigid-limit source non-elementary; the substrate answer is the deformed metric
2026-09-18 05:59Z	0fc13294cf	research(gr-qm): the substrate a_corr is a global field energy; the saturation root resums globally
2026-09-18 07:08Z	b99c392aa1	research(gr-qm): name the two-center Born-Infeld field energy as the open a_corr residual
2026-09-18 14:50Z	674a28b5fd	research(sympy): land a_corr coincidence match derivations and two open verdicts
2026-09-18 15:58Z	caf6916754	research(sympy): land proximity-cap contact-local no-rank verdict for a_corr
2026-09-18 17:19Z	4626c0041a	research: land two-center Born-Infeld field energy R_tail closed-form bracket
2026-09-18 20:06Z	a2ea24ca21	research: prove the two-center Born-Infeld field energy R_tail is non-elementary
2026-09-19 01:27Z	6744384080	research: downgrade R_tail two-center BI structure to a rank-6 upper bound with open obligations
2026-09-19 03:24Z	745293d9c5	fix(physics): derive R_tail trial bounds and correct correlation claims
2026-09-20 12:49Z	de463350ee	fix(gr-qm): reconcile conditional atomic energy and ionization
2026-09-20 23:12Z	82ead9b8d9	fix(gr-qm): consume exact relaxed occupation certificates
2026-09-21 05:09Z	409d83bbff	feat(gr-qm): two-center Born-Infeld minimizer is unique and two-sided bracketed
2026-09-21 14:34Z	35a424b90d	docs(gr-qm): label the lambda-reduction as the frozen superposition energy, not the true R_tail
2026-09-21 15:10Z	826a804f28	feat(gr-qm): validate the M11 true-minimizer route to a_corr - OPEN, blocking antecedent named
2026-10-28 23:16Z	33b095be04†	docs: record acorr derivation history and remaining antecedents
2026-11-01 03:55Z	ef75344b6c	feat(gr-qm): establish selected physical energy order lower bound
2026-11-01 12:06Z	e3b4470853†	feat: derive order17 frozen energy annihilator
2026-11-01 13:13Z	9188e5a275†	feat: classify order17 apparent divisors
2026-11-01 14:03Z	703d37e793†	test: independently check order17 divisor and endpoint data

Appendix C: Premise ledger

Consumed by the definition and the diagnostic numbers: the nonrelativistic Coulomb comparison problem and its $1/Z$ expansion (imported reference model); the published ϵ_2 and per-l increments (literature). Consumed by the static two-centre results: the Born-Infeld energy density (the chain's axiom), unit point sources in static Euclidean space (definitions of the diagnostic) and the finite-energy admissible class (commitment). Registered commitments that bound the conclusions: the M11 identification of the static two-centre energy with a_{corr}, and, for the true large-separation law, the induced-dipole structure. The $E_8 \times E_8$ geometry, its D_8 placement and the particle equations are retained in the framework and are not consumed by these calculations.

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[4] P.-F. Loos and P. M. W. Gill, Correlation energy of two electrons in the high-density limit, J. Chem. Phys. **131**, 241101 (2009).

[5] C. Wang, Rates of convergence of the partial-wave expansion beyond the Kato cusp condition, Phys. Rev. A **88**, 032511 (2013).

[6] C. Schwartz, Importance of angular correlations between atomic electrons, Phys. Rev. **126**, 1015 (1962).

[7] D. P. Carroll, H. J. Silverstone, and R. M. Metzger, Piecewise polynomial configuration interaction natural orbital study of $1s^2$ helium, J. Chem. Phys. **71**, 4142 (1979).

[8] T. Kato, On the eigenfunctions of many-particle systems in quantum mechanics, Commun. Pure Appl. Math. **10**, 151 (1957).